

A Unified Benchmark of Grokking Predictors
in Neural Networks

L2 Norm Predictor

Investigation Report

Phase 2 - Predictor 1 of 9

Prepared by: Jonathan John

M.Sc. Artificial Intelligence, IU Internationale Hochschule
Supervisor: Prof. Dr.-Ing. Sheikh Faisal Rashid

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Summary: This report documents all detection strategies attempted for the L2 Norm grokking predictor, the evaluation criteria used to judge them, and results across three independent training runs on the $(a+b) \bmod 97$ task. No strategy tested so far satisfies all acceptance criteria simultaneously. The findings are reported as a benchmark result, not a blocked task, and a recommendation for how to proceed is given at the end.

1. Objective & Experimental Setup

Objective: evaluate the L2 Norm of model weights as a leading indicator of grokking - i.e. find a rule that reliably fires BEFORE the model's test accuracy jumps, using only the L2 norm history recorded during training. This is the first of 9 predictors in the benchmark's evaluation order (L2 Norm -> Dropout -> Spectral -> AGE -> HTSR Alpha -> Correlation Traps -> Weight-PCA -> Higher-MI -> Commutator Defect).

Task & Architecture

- Task: modular addition $(a + b) \bmod 97$, Nanda et al. style grokking setup.
- Input format: combined token sequence `[a, b, "="]`, predicted from the `"="` position.
- Model: single-head, single-layer Transformer - token + position embedding, Q/K/V self-attention with softmax, residual connection, MLP (4x expansion, ReLU) with residual, linear output head. `d_model = 128`. No LayerNorm.
- Optimizer: AdamW, learning rate `1e-3`, `weight_decay = 1.0` (strong regularization, standard for inducing grokking).
- Data split: 30% train / 70% test (9409 total pairs -> 2822 train / 6587 test).
- Training length: 10,000 epochs, run on CPU (MPS/CUDA unavailable in this environment).
- Grokking criterion: first epoch where test accuracy exceeds 90%.

2. Detection Strategies Attempted (Chronological)

1. Raw rate-of-decline threshold

ABANDONED - the signal was inverted: L2 norm declines FASTEST during early memorization, not during the grok transition. A threshold rule on decline rate cannot distinguish the two phases.

2. Second-derivative inflection (acceleration sign flip)

ABANDONED - raw acceleration was dominated by noise; the rule always fired at the skip-epoch boundary itself rather than a real feature.

3. Double-smoothed inflection

ABANDONED - visually cleaner but reset in favor of a different family of approaches (moving-average crossover) before being fully evaluated.

4. Fast/Slow MA crossover (windows 50/200, on raw L2 norm)

ABANDONED - the naive "first crossover" always fired on early-training noise. Lead times were 3000-5000+ epochs on every run tested - far too early and imprecise to be useful.

5a. MA-of-MA: threshold trigger (fast MA of slow MA, fire at 10x noise floor)

ABANDONED - a fixed multiplier is not robust across runs. On a second training run, two early-training humps (12.8x and 19x the noise floor) both exceeded the threshold before the real late-training signal, causing the rule to fire far too early (epoch 205 vs. grok epoch 4806).

5b. MA-of-MA: first zero-crossing trigger

TESTED ACROSS 3 RUNS - the point where (fast MA of slow MA) - (slow MA) first crosses from positive to negative. Landed in a visually consistent, clean spot on two runs, which motivated adopting it as the active strategy. FAILED on a third run: fired 1884 epochs AFTER the model had already grokked (negative lead time). See Section 4 for full evidence.

3. Formal Criteria for a Valid Trigger

To avoid judging candidate detection rules by eye alone, the following three criteria were established and applied consistently to every strategy tested:

Criterion 1 - Always predictive, never postdictive

The trigger must fire BEFORE grokking on every run, without exception. A rule that sometimes fires after the event is not predicting anything - it is coincidentally correlated some of the time. This is the non-negotiable baseline requirement for calling something a predictor.

Criterion 2 - Consistent, tight relationship to the actual event

The gap between the trigger epoch and the grok epoch must stay small AND consistent as a proportion of the total run length, across different runs with different grok epochs. Landing at a similar ABSOLUTE epoch across runs is not sufficient evidence of a real relationship, since the grok epoch itself moves between runs - a trigger must move WITH it.

Criterion 3 - Clearly above the noise floor

The signal at the trigger point must be large relative to the natural fluctuation seen during quiet, uneventful training (the "noise floor"), not indistinguishable from baseline jitter.

4. Cross-Run Evidence

Three independent training runs were used (training is stochastic - each run has a different random initialization and data shuffling, producing a different grok epoch).

Table 1: First Zero-Crossing Trigger (the adopted strategy)

Run	Grok Epoch	Trigger Epoch	Lead Time	Trigger/Grok Ratio	Verdict
1	5739	2007.8	+3731 (37% of run)	0.350	Weak, but early
2	4806	1688.6	+3117 (31% of run)	0.351	Weak, but early
3	3760	5643.8	-1884 (fires AFTER)	1.501	FAILS Criterion 1

Table 2: Peak-of-Difference (reference candidate, non-causal)

Run	Grok Epoch	Peak Epoch	Lead Time	Gap as % of Run
1	5739	5633.5	+105.5	1.05%
2	4806	4764.0	+42.0	0.42%
3	3760	3742.5	+17.5	0.18%

Note: the peak-based candidate satisfies Criteria 1 and 2 on all three runs tested (always precedes grokking, gap shrinks as a share of the run). However, identifying "the peak" requires seeing the full future curve, so it cannot be used as-is by a live/causal detector - it is included here only as evidence of where the real signal lives, not as a working predictor.

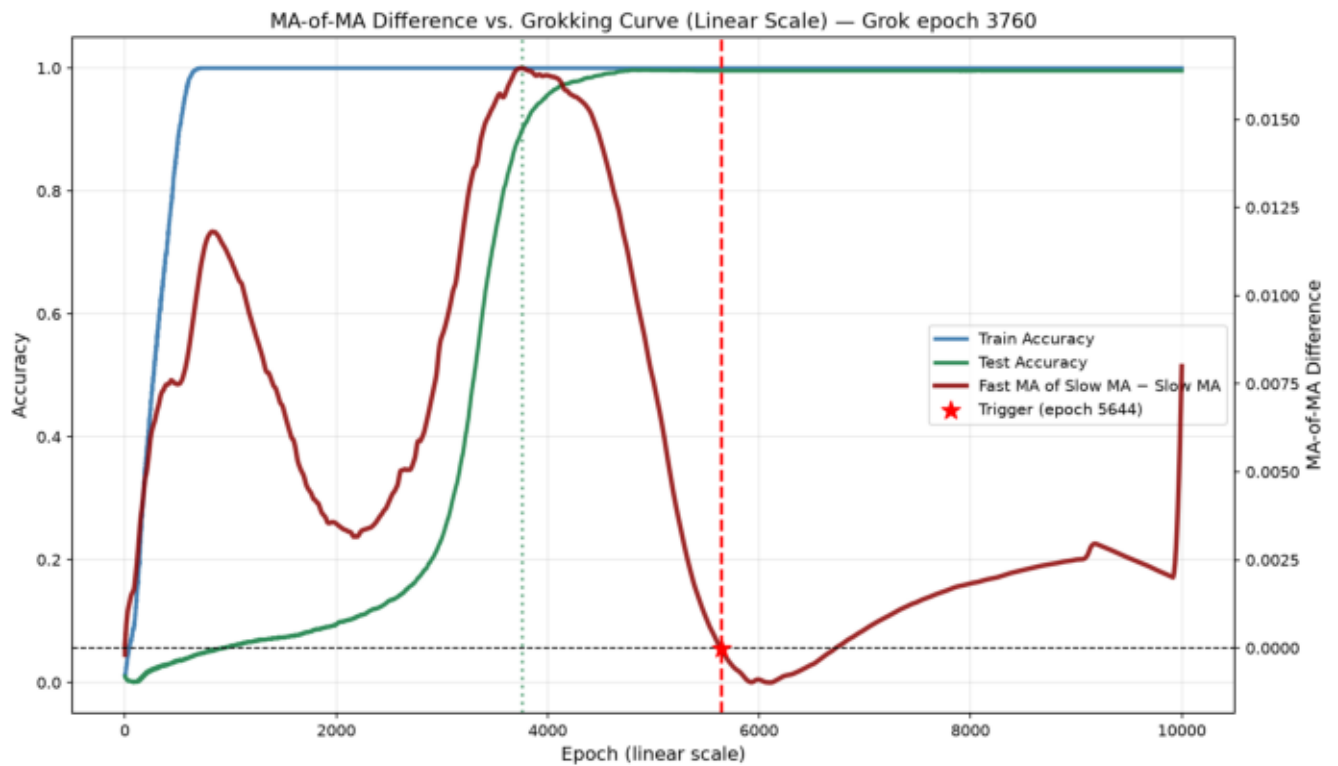


Figure 1: MA-of-MA difference curve overlaid on the grokking curve (Run 3, linear epoch axis). The trigger (red star, epoch 5644) fires well after test accuracy (green) has already plateaued near its final value - the model grokked around epoch 3760, long before the trigger fires.

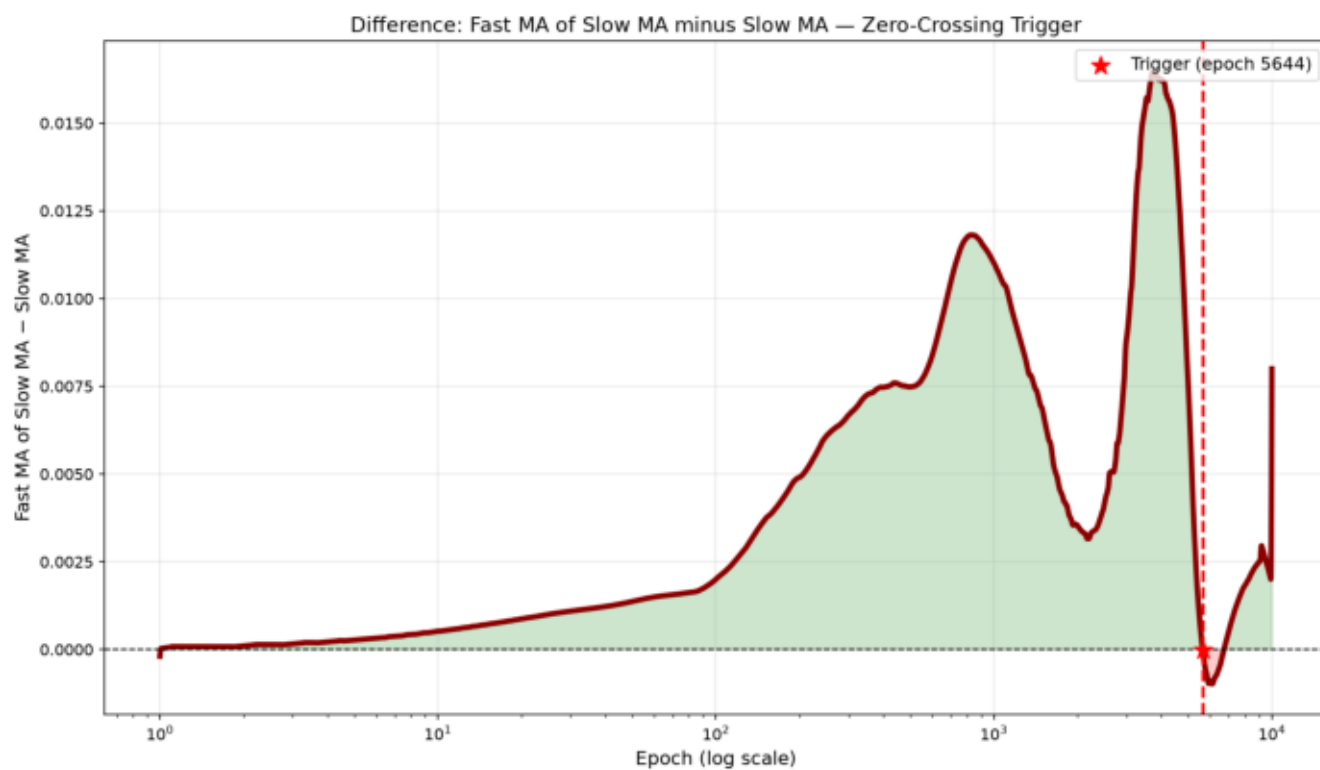


Figure 2: MA-of-MA difference curve (Run 3, log-epoch axis) with the zero-crossing trigger marked. The large, strong peak near the end of training sits much closer to the actual grok epoch than the trigger point does.

5. Side Finding: Test Accuracy Plateau (Run 3)

Run 3 did not reach 100% test accuracy - it plateaued at exactly 99.5446% from approximately epoch 5500 onward, with zero variation through epoch 10,000.

- 99.5446% of 6587 test examples = exactly 6557 correct -> 30 examples permanently wrong, unchanged for 4500+ epochs.
- Training loss frozen near 0.0000834, barely moving - the model has converged to a stable fixed point, not still improving.
- Train accuracy remained a perfect 1.0 throughout.

Likely cause: the model architecture used is intentionally minimal - single attention head,

single layer, and no LayerNorm (Nanda et al.'s original setup typically includes LayerNorm).

Combined with a strong weight_decay of 1.0, this can leave a small, stable set of "hard" input pairs the model never resolves, even after full convergence.

This is a property of the model/training setup, not a bug introduced during the predictor investigation - the training loop and model code were not modified during this work.

Not investigated further this session; flagged as an optional follow-up (identifying which specific (a, b) pairs fail, and whether the failure set is stable across runs).

6. Conclusion & Recommendation

Under every strategy tested, no L2-Norm-derived rule satisfied all three trigger criteria simultaneously across independent runs:

- The zero-crossing trigger looked consistent on two runs, but failed Criterion 1 outright on a third run (fired after grokking, not before) - falsifying it as a reliable predictor.
- The threshold-based trigger failed on robustness grounds - the size of early-training "noise humps" varies enough between runs that no single fixed multiplier reliably separates noise from signal.
- The peak-of-difference candidate is the strongest evidence-backed signal found (always precedes grokking, and its gap shrinks as a share of the run across all 3 runs tested), but it requires reformulation as a causal, rising-edge rule before it can be used as a live detector - this reformulation has not yet been implemented or tested.

Framing for the thesis

The thesis is a benchmark comparing 9 grokking predictors under one protocol, not a proof that all 9 must succeed. A rigorously tested, well-documented negative result for L2 Norm - as recorded here - is a legitimate and defensible outcome for one row of that comparison, provided the failure is due to the signal itself (as shown above) rather than an implementation error.

Recommended next step

Record L2 Norm's result as above (with the peak-based causal variant flagged as untested, not ruled out) and proceed to the next predictor in the evaluation order: Dropout.